REPLICATION KIT

Jobber → Google Calendar Sync

Near real-time Jobber Job Visits on each estimator's Google Calendar, for Routemize availability and job stops.

1. Requirements: accounts, resources, settings, secrets, cost
2. Replication Guide: phase-by-phase setup with verify checks
3. Operations & Troubleshooting: daily tasks, fixes, behaviour
4. Agent Kickoff Prompt: paste into a new Claude agent

Reference build: Cloudflare Worker `jobber-gcal-sync` · Repo `jose24xhr-sys/jobber-gcal-sync`
Prepared October 17, 2026

Jobber → Google Calendar Sync — Requirements

What this builds: a Cloudflare Worker that copies every *scheduled Jobber Job Visit* into the assigned estimator's Google Calendar within seconds. Routemize reads those calendars as availability, and a job stop can be created from each event.

Scope: Job Visits only. Requests, Quotes/Estimates, Assessments, Tasks and Invoices never create calendar events.

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1. Accounts and access

#	Account	Who needs it	What it's used for	Notes
1	**GitHub**	Implementer	Hosts the code; pushes trigger deploys	Private repo. The Claude GitHub app must be installed on the repo if an agent pushes code
2	**Cloudflare**	Implementer	Worker, D1 database, Queues, secrets, logs	The free plan works: Queues include 10,000 ops/day and logs 200K events/day
3	**Cloudflare ↔ GitHub link**	Implementer	Workers Builds auto-deploys from the repo	The Cloudflare account must be linked to the **same** GitHub account that owns the repo
4	**Jobber Developer Center** (developer.getjobber.com)	Implementer	Creates the Jobber app: client ID/secret, scopes, webhooks	A **separate login** from the normal Jobber app
5	**Jobber account** (the customer's)	Jobber **admin** user	Approves the app once ("Allow Access")	A trial account works for testing
6	**Google Cloud** (console.cloud.google.com)	Implementer	Project, Calendar API, service account and JSON key	No billing needed for the Calendar API at this volume
7	**Google Calendar** for each estimator	Each estimator, or their Workspace admin	Shares their calendar with the service account	Only the calendar owner or a Workspace admin can share it
8	**Routemize**	Estimators / admin	Connects each estimator's Google Calendar	Uses the same calendar that receives the Jobber events

2. Tools

- **Browser access** to all of the above. An agent can do almost everything through Claude in Chrome. Exceptions: sign-ins, pasting secrets, approving OAuth screens, and downloading the Google key file are done by the human.
- **Node.js 20+ and npm**, only to run the tests or deploy from a terminal. The normal path deploys from GitHub with no local terminal.
- **Optional:** the Cloudflare MCP connector (D1 queries, creating databases). The repo `jose24xhr-sys/jobber-gcal-sync` holds the complete, tested code: 65 tests.

3. Cloudflare resources

Resource	Name	Binding	Created by
Worker	`jobber-gcal-sync`	—	Workers Builds (import repo)
D1 database	`jobber_gcal_sync`	`DB`	Dashboard or `wrangler d1 create`; then run `migrations/0001_init.sql`
Queue	`jobber-visit-sync`	`SYNC_QUEUE` (producer + consumer)	Dashboard → Queues
Queue (dead letter)	`jobber-visit-sync-dlq`	—	Dashboard → Queues
Cron	daily `17 4 * * *`	—	`wrangler.toml` (prunes old logs)

4. Configuration values

Plain variables (in `wrangler.toml`, safe to commit)

Name	Example	Purpose
`JOBBER_API_VERSION`	`2026-09-25`	Required `X-JOBBER-GRAPHQL-VERSION` header
`JOBBER_REDIRECT_URI`	`https://jobber-gcal-sync.<subdomain>.workers.dev/oauth/jobber/callback`	Must match the Jobber app's Callback URL exactly
`GOOGLE_AUTH_MODE`	`service_account`	Or `oauth` (not recommended)
`DEFAULT_TIMEZONE`	`America/New_York`	Used only for all-day visits
`ALLOWED_TOPICS`	`VISIT_CREATE,VISIT_UPDATE,VISIT_COMPLETE,VISIT_DESTROY,JOB_CREATE,JOB_UPDATE,JOB_DESTROY,APP_DISCONNECT`	Webhook topics acted on
`EVENT_TITLE_PREFIX`	*(empty)* or `[Jobber]`	Optional prefix on event titles

Name	Example	Purpose
`JOBBER_CLIENT_ID`	from the Jobber app	Public identifier
`GOOGLE_SA_CLIENT_EMAIL`	`jobber-sync@<project>.iam.gserviceaccount.com`	Service account identity
`keep_vars = true`	top of `wrangler.toml`	Stops Git deploys from deleting variables added in the dashboard

Secrets (Cloudflare → Worker → Settings → Variables and Secrets → type Secret)

Name	Where it comes from	Format
`JOBBER_CLIENT_SECRET`	Jobber app → Authorization section	As shown. It also verifies webhook signatures
`GOOGLE_SA_PRIVATE_KEY`	The service-account JSON key file	Paste the **one-line** `"private_key": "-----BEGIN…\n…-----END PRIVATE KEY-----\n"` text from the JSON. **Do not paste the multi-line PEM**: the dashboard field keeps only the first line
`TOKEN_ENCRYPTION_KEY`	You generate it	Ideally `openssl rand -base64 32`; any long passphrase also works
`ADMIN_API_TOKEN`	You generate it	Long random string; it protects `/admin` and `/oauth/jobber/start`

Never commit secrets, put them in URLs, or paste them into chat.

5. Jobber app settings

Setting	Value
App name / developer	e.g. "Routemize Calendar Sync" / "Routemize"
Callback URL	`https://jobber-gcal-sync.<subdomain>.workers.dev/oauth/jobber/callback`
Short description	Required to save
Scopes (read only)	**Clients**, **Jobs**, **Scheduled Items**, **Users** — no Write boxes
Webhooks (all → `https://jobber-gcal-sync.<subdomain>.workers.dev/webhooks/jobber`)	`VISIT_CREATE`, `VISIT_UPDATE`, `VISIT_COMPLETE`, `VISIT_DESTROY`, `JOB_CREATE`, `JOB_UPDATE`, `JOB_DESTROY`, `APP_DISCONNECT`
Refresh token rotation	**ON** (default)

6. Google settings

Item	Value
Project	e.g. `jobber-gcal-sync`
API enabled	Google Calendar API (`calendar-json.googleapis.com`)
Service account	e.g. `jobber-sync` — **no project roles**
Key	JSON (Keys → Add key → Create new key → JSON)
OAuth scope used	`https://www.googleapis.com/auth/calendar.events` only
Per-estimator calendar sharing	Share with the service account email → **"Make changes (see private events as free/busy)"**

7. Information to collect from the business

- ☐ List of estimators/techs who should get Jobber visits on a calendar
- ☐ For each: **Jobber user** (name) → **Google Calendar ID**, usually their email; it's under Calendar settings → Integrate calendar
- ☐ Confirmation that each estimator has shared their calendar with the service account
- ☐ Business timezone (for all-day visits)
- ☐ Which Jobber admin will click "Allow Access"

8. Hosting model and cost

Two ways to offer it to other businesses

Model	How it works	Limits
Hosted (one Worker, many businesses)	Each business's Jobber admin opens `/oauth/jobber/start` and clicks Allow Access. Their estimators share calendars with your service account. The code is already multi-account	A custom Jobber app can connect to **5 paying Jobber accounts** at most; beyond that it must be published in the Jobber App Marketplace (review, logo, Terms of Service and Privacy Policy URLs). You hold customers' tokens and client data, so have a privacy policy
Self-hosted (one copy per business)	The business follows this guide with its own Cloudflare, Google and Jobber app	No 5-account limit; costs and data stay with the business

Running cost (prices checked Oct 2026)

Service	Cost	Notes
Cloudflare	**$0** on the free plan	Free plan: 100,000 Worker requests/day, 10,000 Queue operations/day, D1 100,000 rows written/day + 5 GB. One visit change ≈ 2 webhooks ≈ 2 requests + ~6 queue ops, so about 1,500 visit changes/day fit free. Paid: **$5/month** (10M requests + 1M queue ops/month included)

Service	Cost	Notes
Google Calendar API	**$0**	No per-call charge; daily quotas are far above this volume. No billing account required
Claude	**$0 to run**	The Worker never calls Claude; Claude was only used to build it
Jobber	**$0** for API use	The business needs its normal Jobber subscription

Jobber → Google Calendar Sync — Replication Guide

A step-by-step playbook for a person or a Claude agent to stand up a fresh copy of this integration for a new business. Every phase ends with a **Verify** check, so don't start a phase until the one before it passes.

Legend: AGENT = an agent can do it (browser or connector) · HUMAN = the human must do it (sign-in, secret, approval, download)

```
Jobber (visit created/edited/deleted)
    │  webhook: topic + IDs only, HMAC-signed
    ▼
Cloudflare Worker  /webhooks/jobber ── verify signature ── enqueue ── 200 OK (< 1 s)
    │
    ▼  Queue consumer (retries, dead-letter queue)
Jobber GraphQL: visit(id) + job + client + property + assignedUsers
    │  validate: Visit on a Job, scheduled, assigned
    ▼
Google Calendar API (service account) → event on each assignee's calendar
    │
    ▼
Routemize reads the estimator's calendar → availability blocked → job stop can be created
```

Phase 0 — Collect inputs HUMAN

Use the checklist in `01-REQUIREMENTS.md §7`. Decide names now:

Placeholder	Example used in the reference build
`<GITHUB_OWNER>/<REPO>`	`jose24xhr-sys/jobber-gcal-sync`
`<CF_SUBDOMAIN>`	`joseramirez` (from Cloudflare → Workers → Subdomain)
`<WORKER_URL>`	`https://jobber-gcal-sync.<CF_SUBDOMAIN>.workers.dev`
`<GCP_PROJECT>`	`jobber-gcal-sync`
`<SA_EMAIL>`	`jobber-sync@<GCP_PROJECT>.iam.gserviceaccount.com`

Agent rules: never type passwords, API keys or secrets into any field, and never put them in URLs or chat. Ask the human to paste them. Ask before any OAuth/app-install approval, calendar sharing, or file download.

Phase 1 — Code repository

1. AGENT Create a **private** repo `<REPO>` (GitHub → New repository; no README/gitignore/license).
2. AGENT Copy the reference code into it. Either copy the source repo, or push the `jobber-gcal-sync.zip` contents. If an agent pushes from its own workspace, its GitHub app must have access. The safe way: install the **Claude** GitHub app with *Only select repositories* → *`<REPO>`* (HUMAN approves).
3. AGENT Keep `wrangler.toml` placeholders for now; Phase 3 fills them.

Verify: the repo shows `src/`, `test/`, `migrations/`, `wrangler.toml`, `README.md`. Optional: `npm ci && npm test` → **65 passed**.

Phase 2 — Cloudflare data resources

1. AGENT Pick the right Cloudflare account. If there are several, use the one the business wants to own the Worker.
2. AGENT **D1:** create database `jobber_gcal_sync` (location: closest to the business, e.g. ENAM). Copy its **database ID**.
3. AGENT **Schema:** run all of `migrations/0001_init.sql` against it, through the D1 Console in the dashboard or the Cloudflare connector's `d1_database_query`. It creates 5 tables: `jobber_accounts`, `oauth_states`, `user_calendar_map`, `visit_events`, `sync_log`.
4. AGENT **Queues:** create `jobber-visit-sync` and `jobber-visit-sync-dlq` (Workers → Queues → Create Queue; jurisdiction None).
5. AGENT Read the account's workers.dev subdomain (Workers → *Subdomain* page). **Do not change it**: that would move every Worker on the account.

Verify: D1 lists 5 tables; Queues lists 2 queues (status "Inactive" is normal until deploy).

Phase 3 — Configure and deploy the Worker

1. AGENT Edit `wrangler.toml`:
   - `database_id = "<D1 ID>"`
   - `JOBBER_REDIRECT_URI = "<WORKER_URL>/oauth/jobber/callback"`
   - `DEFAULT_TIMEZONE` = the business timezone
   - keep `keep_vars = true` at the top
   - Commit and push.
2. AGENT Cloudflare → Workers & Pages → **Create** → **Continue with GitHub** → pick `<REPO>` → deploy. Defaults are fine: deploy command `npx wrangler deploy`, root `/`.
   - If the repo doesn't appear, the Cloudflare account is linked to a different GitHub account. Choose **Add GitHub account** and HUMAN authorizes Cloudflare's GitHub app for `<GITHUB_OWNER>`. The approval opens in a pop-up the agent can't see, so ask the human to finish it.
3. AGENT Wait for the build: Initializing → Cloning → Installing → Deploying, all green.

Verify: `GET <WORKER_URL>/health` → `ok`.

From now on, **every push to `main` redeploys automatically**, in about 1 minute.

Phase 4 — Jobber app

1. HUMAN Log in at developer.getjobber.com (separate from the normal Jobber login).
2. AGENT Manage Apps → **New**. Fill in:
   - App name, Developer name, **Callback URL** `<WORKER_URL>/oauth/jobber/callback`, **Short description** (required).
   - **Scopes**: toggle ON **Clients**, **Jobs**, **Scheduled Items**, **Users**. Leave **Write** unchecked.
   - **Webhooks**: add 8 rows, all with URL `<WORKER_URL>/webhooks/jobber`: `VISIT_CREATE`, `VISIT_UPDATE`, `VISIT_COMPLETE`, `VISIT_DESTROY`, `JOB_CREATE`, `JOB_UPDATE`, `JOB_DESTROY`, `APP_DISCONNECT`.
     - NOTE The topic picker is a type-ahead. **Click the option**; pressing Enter selects the first item in the full list (`APP_CONNECT`).
   - Refresh token rotation: leave **ON**.
   - **Save App**.
3. AGENT Read back all 8 topic/URL pairs (e.g. from the form's input values) before saving.

Verify: "App updated" toast. The Authorization section shows Client ID and Client Secret.

Phase 5 — Google Cloud

1. AGENT console.cloud.google.com → New project `<GCP_PROJECT>` (no organization is fine).
2. AGENT APIs & Services → Library → **Google Calendar API** → **Enable**.
3. AGENT IAM & Admin → Service accounts → **Create**: name `jobber-sync`. **Skip roles** and close the role picker; none are needed.
4. HUMAN Service account → **Keys** → Add key → Create new key → **JSON**. Downloads a file; keep it private.
   - If key creation is blocked by an org policy (`iam.disableServiceAccountKeyCreation`), a Google admin must allow it for this project.

Verify: the API page shows **Status: Enabled**; the service accounts list shows `<SA_EMAIL>`.

Phase 6 — Secrets and variables

HUMAN Cloudflare → Workers & Pages → `jobber-gcal-sync` → **Settings → Variables and Secrets → + Add** (type **Secret** for all four):

Name	Value
`JOBBER_CLIENT_SECRET`	From the Jobber app

Name	Value
`GOOGLE_SA_PRIVATE_KEY`	In the JSON file, the single line `"private_key": "-----BEGIN PRIVATE KEY-----\nMII…\n-----END PRIVATE KEY-----\n",` copied **whole**. Extra quotes or the label are fine; the code cleans them. Never paste the multi-line PEM: the field keeps only the first 64 characters
`TOKEN_ENCRYPTION_KEY`	Long random value. **Set it once, before connecting Jobber.** Changing it later means reconnecting
`ADMIN_API_TOKEN`	Long random value; save it in a password manager

AGENT Put `JOBBER_CLIENT_ID` and `GOOGLE_SA_CLIENT_EMAIL` in `wrangler.toml` `[vars]` (not secret) and push.

Verify (agent): the Settings page lists all 6 names. `TOKEN_ENCRYPTION_KEY`, `JOBBER_CLIENT_SECRET`, `GOOGLE_SA_PRIVATE_KEY` and `ADMIN_API_TOKEN` must show **"Value encrypted"**. If any shows a plain value, ask the human to edit it and switch the type to **Secret**.

Phase 7 — Connect Jobber

1. **HUMAN** Open `<WORKER_URL>/oauth/jobber/start` → paste `ADMIN_API_TOKEN` in the password box → **Continue**.
2. **HUMAN** In Jobber's consent screen, as a Jobber **admin**: **Allow Access**.
3. The success page shows **"Connected Jobber account `<name>`"** and the **Account ID** (e.g. `Z2lkOi8vSm9iYmVyL0FjY291bnQv…`).

Verify (agent, via D1):

```
SELECT account_id, account_name, status FROM jobber_accounts;   -- status = active
```

Phase 8 — Share calendars and map estimators

1. **HUMAN** Each estimator, or the Workspace admin, opens Google Calendar → Settings → *their calendar* → **Shared with → Add people and groups** → `<SA_EMAIL>` → **Make changes (see private events as free/busy)** → Send.
   - An agent can do this only for calendars owned by the Google account signed in to the browser, and only after the human agrees.
2. **AGENT** / **HUMAN** Open `<WORKER_URL>/admin` → enter `ADMIN_API_TOKEN`. The page lists every Jobber team member. Type each estimator's **Calendar ID** (usually their email) → **Save mappings**. Leave a row blank to not sync that person.
   - Alternative (agent, no token needed): insert into D1 directly. The API user ID = base64 of `gid://Jobber/User/<n>`. `<n>` comes from Jobber web → Settings → Manage Team, whose link `/manage_team/<base64 n>` holds the number base64-encoded.

```
INSERT INTO user_calendar_map (account_id, jobber_user_id, google_calendar_id, label, active,
updated_at)
VALUES ('<ACCOUNT_ID>', '<USER_ENCODED_ID>', 'estimator@company.com', 'Estimator name', 1,
strftime('%s','now')*1000)
ON CONFLICT(account_id, jobber_user_id) DO UPDATE SET google_calendar_id =
excluded.google_calendar_id, active = 1;
```

3. HUMAN Connect the same calendars in **Routemize** for each estimator, if they aren't already.

Verify: `/admin` shows the calendar next to each estimator.

Phase 9 — End-to-end test

1. AGENT In Jobber: **Create → Job** for an existing client, title `TEST - Calendar sync check`, one visit today or tomorrow with start and end times, **assigned** to a mapped estimator, instructions "Safe to delete" → **Save Job**.
2. AGENT Within ~10 s, `/admin` → *Recent activity* shows `created VISIT_CREATE`. A `noop` right after is normal: Jobber sends a second webhook, and it's de-duplicated.
3. AGENT Google Calendar (that estimator) shows "**`<Client> - <Job title>`**" at the right time, with the property address as location and client, job #, instructions and the Jobber link in the description.
4. Then test the rest:

Do in Jobber	Expect in Google Calendar
Reschedule the visit	Same event moves (no duplicate)
Reassign to another mapped person	Event leaves calendar A, appears on calendar B
Assign two people	One event on each calendar
Unschedule the visit	Event removed
Delete the visit / job	Event removed
Create a Request or Quote	Nothing

5. HUMAN In Routemize: the slot is blocked and a job stop can be created from the event.
6. AGENT Delete the TEST job when done; its event is removed automatically.

If nothing appears: see `03-OPERATIONS.md → Troubleshooting`. The Cloudflare **Observability → Events** log shows the exact error per message.

Done criteria

- ☐ `/health` returns `ok`
- ☐ 4 secrets show "Value encrypted"; `keep_vars = true` is in `wrangler.toml`
- ☐ `jobber_accounts.status = active`
- ☐ Every estimator is mapped on `/admin`, and their calendar is shared with `<SA_EMAIL>`

- ☐ Create, reschedule, reassign and delete all behave as in the Phase 9 table
- ☐ The Requests/Quotes test produces no events
- ☐ TEST job deleted

Jobber → Google Calendar Sync — Operations & Troubleshooting

Day-to-day tasks

Add a new estimator

1. In Jobber, add them as a team member (Settings → Manage Team).
2. They share their Google Calendar with `<SA_EMAIL>` using **Make changes (see private events as free/busy)**.
3. `<WORKER_URL>/admin` → enter the admin token → type their Calendar ID next to their name → **Save mappings**.
4. Connect the same calendar in Routemize.

Their visits sync from the next change onward. To backfill an existing visit, edit it in Jobber (any save fires a webhook).

Remove or pause an estimator

On `/admin`, clear their Calendar ID and save. Events already on their calendar stay until each visit changes or is deleted. Then remove the calendar share.

Move an estimator to a new calendar

Change the Calendar ID on `/admin`. On each visit's next change, the event moves from the old calendar to the new one.

Where to look

Need	Place
Recent syncs per visit (created/updated/deleted/skipped/error)	`<WORKER_URL>/admin` → Recent activity, or D1 `sync_log`
Full request/queue logs with error text	Cloudflare → Worker → **Observability → Events**
Failed messages after 8 retries	Queue `jobber-visit-sync-dlq`
Which calendar/event a visit maps to	D1 `visit_events` (or JSON `GET /admin/visits?accountId=…&visitId=…` with `Authorization: Bearer <ADMIN_API_TOKEN>`)
Re-run one visit	JSON `POST /admin/resync {"accountId","visitId"}` with the bearer token, or just edit the visit in Jobber

Useful D1 queries

```
-- Last 20 sync actions
SELECT datetime(at/1000,'unixepoch') AS utc, topic, action, detail FROM sync_log ORDER BY id DESC
LIMIT 20;
-- Connection health
SELECT account_name, status, datetime(access_expires_at/1000,'unixepoch') AS token_expires FROM
jobber_accounts;
-- Mappings
SELECT label, jobber_user_id, google_calendar_id, active FROM user_calendar_map;
```

Troubleshooting (every row was hit or prevented during the reference build)

Symptom	Cause	Fix
`/oauth/jobber/start?key=…` says **unauthorized**	The placeholder was left in the link, or the token has `# & + $` characters that break inside URLs	Open `/oauth/jobber/start` with no `?key` and type the token into the password box
Log: `Invalid PKCS8 input`; diagnostic shows `base64Length: 64`, `hasBeginMarker: false`	The multi-line PEM was pasted into the single-line secret field, so only the first line was kept	Re-paste `GOOGLE_SA_PRIVATE_KEY` using the one-line `"private_key": "…\n…"` text from the JSON file. Queued visits sync automatically once fixed
Log: `TOKEN_ENCRYPTION_KEY must be 32 bytes` (old builds)	A passphrase was used instead of 32-byte base64	Current code accepts any passphrase. Make sure it's type **Secret**
Dashboard variables disappear after a GitHub push	Wrangler replaces variables not in `wrangler.toml`	`keep_vars = true` (already set); put non-secret values in `wrangler.toml`
Cloudflare "Import repository" doesn't list the repo	The Cloudflare account is linked to a different GitHub account	**Add GitHub account** → authorize for the repo owner (human finishes the pop-up)
Agent can't push: "repository not found / no access"	The Claude GitHub app isn't installed on that private repo	Install the Claude app → *Only select repositories* → the repo
Jobber webhook topic saved as `APP_CONNECT`	Pressed Enter in the type-ahead	Click the option instead; read back all rows before saving

Symptom	Cause	Fix
Nothing in `sync_log` and no "webhook queued" in logs	Webhooks not reaching the Worker: wrong URL, app not saved, or account not connected	Check the Jobber app's webhook URLs = `<WORKER_URL>/webhooks/jobber`; `jobber_accounts.status = active`
Log: `Rejected webhook: invalid or missing signature`	`JOBBER_CLIENT_SECRET` doesn't match the app (or the secret was regenerated)	Re-paste the secret from the Jobber app
`sync_log`: `No calendar mapping for Jobber user …`	The assignee isn't mapped	Map them on `/admin`
Error: Google Calendar **403/404** on a calendar ID	That calendar isn't shared with `<SA_EMAIL>`, or the ID is wrong	Share with "Make changes…"; check the Calendar ID under Settings → Integrate calendar
`jobber_accounts.status = reauth_required`	Refresh token invalid (app disconnected, secret regenerated, scopes changed)	Re-run `/oauth/jobber/start`
Two identical events on one calendar	Two assignees are mapped to the **same** calendar (normal in testing)	Map each person to their own calendar
All-day visit on the wrong date	`DEFAULT_TIMEZONE` doesn't match the business	Set it in `wrangler.toml` and push
`Jobber API version warning` in logs	The pinned `JOBBER_API_VERSION` is nearing end of support	Check developer.getjobber.com/docs/changelog, bump the version, push

Behaviour reference

- **Idempotent:** the event ID is derived from (account, visit, user). Repeated or duplicate webhooks never create duplicates. An unchanged visit causes zero Google writes (`noop`).
- **New jobs:** `JOB_CREATE` fetches every visit of the new job (up to 100) and syncs each one. Jobber also sends `VISIT_CREATE` per visit, so this is a safety net, and the duplicates are no-ops.
- **Deletes survive the visit being gone:** the visit→event mapping is stored before the event is created.
- **Retries:** network errors, 429 and 5xx are retried inline, then by the queue with backoff (15 s → 15 min, 8 attempts), then sent to the DLQ. Config errors that won't fix themselves (calendar not shared) are logged and not retried. A bad Google key **is** retried, so visits sync once the key is fixed.

- **Token safety:** Jobber refresh-token rotation is handled with a single-refresher lock. Tokens are AES-GCM encrypted in D1 and never logged.
- **Events:** busy (opaque), no reminders, no invitation emails. Title = Jobber visit title (Jobber defaults it to "Client - Job title"). Location = full property address. Description = client, job #, instructions, Jobber link.

Verified live in the reference build (Oct 17, 2026)

- Webhook received → HMAC verified → queued → Jobber GraphQL fetch → Google event created on the mapped calendar, with correct time and address
- Duplicate follow-up webhook recognized (`noop`)
- Reschedule, reassign, unschedule, delete and Request/Quote filtering are covered by the automated test suite (65 tests); run the Phase 9 table on each new install.

Agent kickoff prompt

Paste the block below into a new Claude agent session. Attach `01-REQUIREMENTS.md`, `02-REPLICATION-GUIDE.md`, `03-OPERATIONS.md` (or the PDFs) and give it access to the code repo.

```
You are setting up the "Jobber Job Visits → Google Calendar" integration for a new business.
It is a Cloudflare Worker (code is complete and tested — do not rewrite it) that mirrors
scheduled Jobber Job Visits into each assigned estimator's Google Calendar within seconds,
so Routemize sees them as availability and can create job stops from them.

Source code: GitHub repo <SOURCE_REPO> (copy it into a new private repo for this business).
Follow 02-REPLICATION-GUIDE.md phase by phase. Do not start a phase until the previous
phase's "Verify" check passes. Use 01-REQUIREMENTS.md for every name, scope, topic and
setting. Use 03-OPERATIONS.md when anything fails — check Cloudflare Observability logs
and the D1 sync_log before changing code.

Business inputs:
- Business / Jobber account name: <NAME>
- Cloudflare account to use: <CF_ACCOUNT_EMAIL>
- GitHub owner for the new repo: <GITHUB_OWNER>
- Business timezone: <IANA_TZ, e.g. America/New_York>
- Estimators (Jobber name → Google Calendar ID):
  - <Name> → <calendar@company.com>
  - <Name> → <calendar@company.com>

Rules:
- Never type passwords, API keys, client secrets, private keys or tokens into any field,
  URL or chat. Ask me to paste them. Ask me before: installing/authorizing any app,
  sharing a calendar, downloading a file, or creating records in a live Jobber account.
- Sign-ins are done by me. If a pop-up window opens, ask me to finish it.
- Jobber webhook topic picker: click the option, never press Enter.
- GOOGLE_SA_PRIVATE_KEY must be pasted as the one-line JSON "private_key" value.
- Keep keep_vars = true in wrangler.toml; non-secret values go in [vars].
- Finish with the Phase 9 test table and the "Done criteria" checklist, then report
  what passed, what is pending (e.g. estimators who haven't shared calendars), and
  delete the TEST job.
```